

Some Provocation Tests

Provocative testing is intended to identify the postures, motions, and loads that cause discomfort or pain. They are used to guide the design of pain-free therapeutic exercise and avoidance strategies to remove the cause. The following list comprises some that we have designed or quantified (or both). Because they may mimic the mechanism of injury, it is up to you, the clinician, to set the intensity so that it is sufficient only to provoke discomfort and not to risk further injury.

Some video clips are provided in the online video that accompanies this text, I have also made a DVD of the tests and patients' reactions to them. The many nuances for diagnosis and treatment decisions are demonstrated in the DVD Assessment and Therapeutic Exercise. If you are seeking to become an expert diagnostician, I encourage you to view this unique footage, which is available at www.backfitpro.com.

Compression Tests

Each compression test is well suited to establish compressive tolerance in specific postures and to provide clues for the working diagnostic hypothesis. In compression tests, patients generate their own loads. Caution them not to crush themselves, but simply to perceive whether the pain is worsened.

Heel Drop Test

Schmorl's nodes and fractured end plates result from compressive overload. Annulus damage typically does not compromise compression tolerance until the disc is flexed. The heel drop test can cause pain or feel uncomfortable in those with pain-provoked compression in an upright and neutral spine. True periosteum pain is often described as local and boring. The dynamic load peak is usually around 2.5 times body weight as seen on a force plate (registering total body weight). A positive pain response would strengthen a hypothesis of recent end-plate damage, or perhaps compromise to the vertebral body (e.g., cancellous bone damage, osteoporotic wedge fracture). If the patient adopts a more extended posture, and the compression

is repeated, provoked pain may suggest more posterior bony damage such as to the facet joints or neural arch.





The standing heel drop test is performed with the patient standing relaxed (a), then rising onto the balls of the feet. The patient then drops down to a flat foot, causing a rapid compressive loading of the spine (b). Although the test is conducted cautiously, repeated trials when the patient is pain free may be

more aggressive until a rigid drop produces a lumbar compressive magnitude up to about 2.5 times the body weight. Then bracing is performed to evaluate whether it is an effective pain-reducing strategy (cued here with the instruction to harden the abdominals out laterally) (c).

Two patients come to mind as examples of responses to the heel drop test. The first had a T9 wedge fracture, and upon heel drop compression, local spine pain (lower thoracic) was coupled with rectus abdominis sensations of numbness (rectus is innervated from this spinal level). The second patient had prosthetic discs at L5 and L4 and a flattened disc at T10. Once again, the heel drop produced local lumbar pain together with rectus abdominis perceptions, which is innervated from the T10 level, revealing that multiple levels were causing the multiple symptoms in the patient.

Other techniques that provide insight during the heel drop test are tuning the brace and changing cervical posture. If pain is provoked during the uncoached heel drop, instruct the patient to brace the abdominals and repeat. Adjust the intensity of the brace to see whether pain changes. If the pain worsens, this indicates that the spine cannot tolerate the extra compression resulting from the brace. Instead, try a shoulder-bracing strategy in which the pectorals and latissimus dorsi muscles cocontract, pulling the shoulders down into the antishrug. Test whether this reduces the pain.

If radiating symptoms are exacerbated with the heel drop, follow up by adjusting the cervical posture to influence neural tension. For example, if the right great toe goes numb with the heel drop, have the patient bend the cervical spine to the right and repeat. Then bend the cervical spine to the left. Changing pain patterns define the neural tensions and spine postures that relieve and exacerbate pain.

Seated Compression Tests

Seated compression tests reveal whether compression intolerance is modulated by posture. For example, posterior disc herniations generally tolerate compression better when the spine is in a neutral posture, yet these patients feel very uneasy when the spine is flexed. This makes perfect sense considering that flexion creates hydraulic pressure on a posterior bulge. The seated compression test is helpful in such cases. Have the patient sit upright, grab the stool seat pan and pull up; then repeat with the spine (not the neck) flexed in a slouched sitting posture.









The photos show (a) seated compression with the spine in an upright posture; (b) then compression is repeated in a slouched posture. The patient pulls upward on the stool seat and repeats for each variation of the test. Have the patient adjust the spine posture, lift the rib cage, and repeat. This may create pain at the thoracolumbar junction. Alternatively, have her flex the hips to align the spine vertically and repeat the test. Then have her flex the neck to see whether neural tensions modify the pain sensitivity (c). Then laterally flex the neck to see if radiating pain occurs (d). If pain occurs, rest whether it is relieved by extending the neck but maintaining a slouched torso (e). This will reveal the link between pain and neural tension. Find the posture that affords the highest tolerance, and teach that posture and strategy to the patient.

Also be observant during the standing up motion. Pain may be from the discs and SI joints, but rarely from the facet joints (Young, Aprill, and Laslett, 2003).

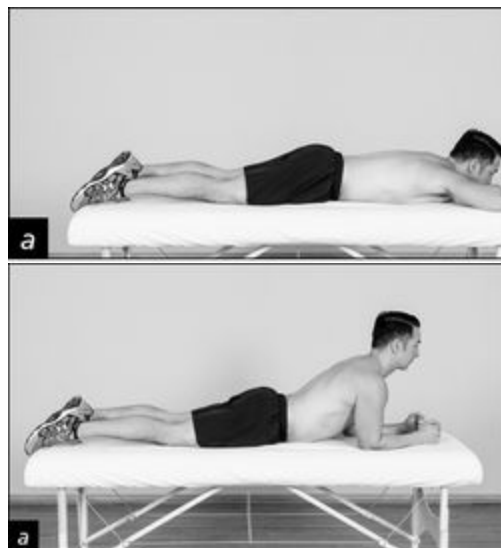
Extension Tests

Spine extension as a provocative test can be interesting, because the mechanics involve several structures. For example, a prone, relaxed extension (some McKenzie postures) may relieve pain, suggesting that the pain was discogenic. Although for years the mechanism was thought to be actual migration of the nucleus anteriorly, recent work has shown a more interesting phenomenon. As the annulus slowly and progressively breaches, layer by layer, nuclear material infiltrates the annulus. Prone extension has been shown to wring out the annulus and squeeze the nucleus fluids back into the annulus in some patients (Scannell and McGill, 2009). But this also depends on the type of annulus damage, its location, disc shape, and the disc height remaining. Obviously, a more posteriorly located partial herniation should respond to this hydraulic mechanism. Sometimes the delaminations of the layers in the annulus travel around to the lateral, and even anterior, portions of the annulus. In these cases, extension may drive more nuclear material anteriorly following the path of delamination. Clinically, a side bend (known as the side glide to McKenzie-based clinicians) is effective for some patients. We speculate that these may be the ones with lateral delaminations. The side glide is then followed by extension, which, once again, is effective for some. The nuclear material may be then directed back to the central nucleus or may be driven back to the lateral rent.

Prone Extension Test

The patient begins in a relaxed standing posture, and you ask about current pain and the general feeling about the back. Then the patient lies prone, adopting one of the following three levels of the extension posture progression. (Decide which is suitable based on the patient's flexibility and symptoms. More extension is not better; simply find what is most tolerable).

- (a) Prone with the arms relaxed, hands are flat and stacked under the chin
- (b) Prone and supported on the elbows
- (c) Prone with elbow support but with cervical flexion (this reduces pain in a patient with an underhooked nerve root)





While in position a during the prone extension test, the pain mechanism can be probed. Begin with the forehead on the hands (d). If the patient experiences discomfort, push the eyebrows down to see if this stabilization maneuver helps relieve the discomfort. Then try the chin on the hands (e), then chin on one fist (f), then the chin on two stacked fists (g).





If the patient cannot tolerate one of these postures, you can assume that simple disc disturbances are not the source of troubles, or at least not the sole source. Many with uncomplicated posterior disc lesions, or herniations, find relief in these postures (Young, Aprill, and Laslett, 2003). The patient then returns to the standing posture. A patient who feels more stable or has less pain compared to the moments prior to lying prone is classified as posterior discogenic.

A therapy that greatly assists these patients is learning to lie prone in their relieving posture and then inhaling. They then exhale while consciously feeling the lumbar region fall toward the table (a).

Patients who are posterior discogenic should avoid flexion stretches and a flexed spine under load when performing exercise therapy. Those who feel worse upon standing could have a host of other possible conditions but will probably experience better progress by adopting a neutral to slightly flexed lumbar posture for exercise therapy.

Traction is then applied in the prone position first as a test, then subsequently as a treatment if it relieves pain. This provides greater relief for some patients, but exacerbates the symptoms of others. Traction applied to the ankles may help or hinder pain relief. The more unstable the joint is, the more likely traction will be to exacerbate the pain. Clinical tricks can be added throughout to help you understand the pain trigger and the associated avoidance strategy. For example, if pain is triggered during prone traction, cue the patient to push one eyebrow into the table and see whether the pain is relieved. Then have the patient turn the head and try the other eyebrow. This added muscular stiffness reduces symptoms in many patients, thus identifying a pain-reducing strategy once a specific trigger has been identified.



Postures to relieve pain for the posterior discogenic patient: Lying prone, as in panel a of the prone extension test, the patient inhales and then exhales focusing on allowing the lumbar spine to fall to the table. This gentle lumbar extension accelerates the stress relief for discs that respond well to prone extension. This does not cause the irritation of the facet joints that eventually occurs in those performing the floppy push-up, which our work has shown should be avoided. Static prone postures are sufficient. Traction applied to the ankles may help or hinder pain relief. The more unstable the joint is, the more likely traction will be to exacerbate the pain. (b) Sometimes it is more effective to lift the knees from the table. Then a small wiggle of the legs side-to-side at a rate of two cycles per second is about the most effective vacuuming technique to reduce a posterior disc bulge. However, as this first is a test, pain with the wiggle suggests instability (see the test for spondylolisthesis in the next section of this chapter).

Tests for Determining Suitability for ROM Training and Stretching

The McKenzie extension approach is well known for treating acute discogenic troubles. But the extension postures can also be used as a test to

identify those with posterior disc troubles. In most cases, these posterior discogenic patients should never flex the lumbar spine when it is under substantial loading. See the preceding section prone extension test.

Spondylolisthesis Test

Most spondylolisthesis patients' pain is exacerbated by extension postures and anterior shear loads. In this test the patient lies prone. Pure traction (about 20 lb or 90 N) is applied via the ankles. Then the clinician raises the legs until the knees just clear the table. Back pain and sometimes radiating symptoms occur if the joint is unstable and a pain generator. Then a slow (1 cycle every 2 seconds) side-to-side shake may be applied. Pain provocation for spondylolisthesis begins with the patient prone. Traction loads of approximately 20 lb are applied (10 lb per leg); then the ankles are raised until the knees clear the table. Pain indicates a positive test. Conversely, this may provide relief suggesting that the disc, not spondylolisthesis, is the pain source. A slow side-to-side shake helps quantify the stability of the joint: pain is linked to instability, and relief is linked to stability. The more unstable the joint and listhesis is, the greater the pain is. Conversely, a more stable joint will enjoy the accelerated return of a posterior disc bulge with pain reduction that this technique creates.

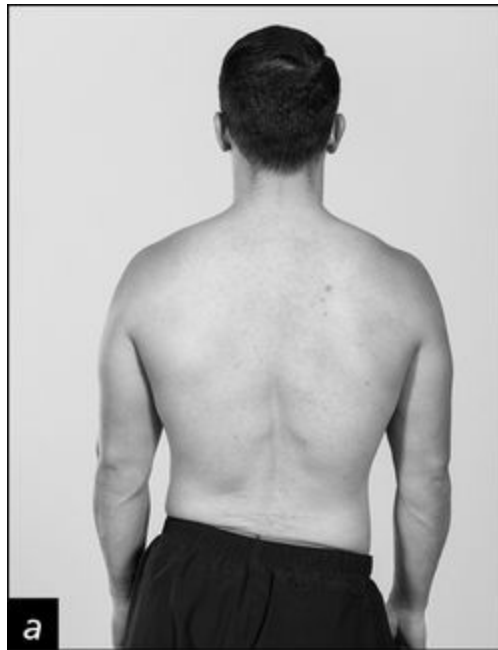


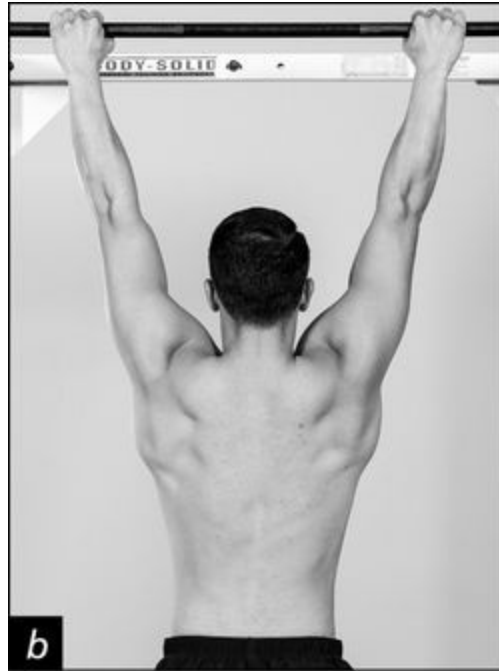
Scoliosis Test

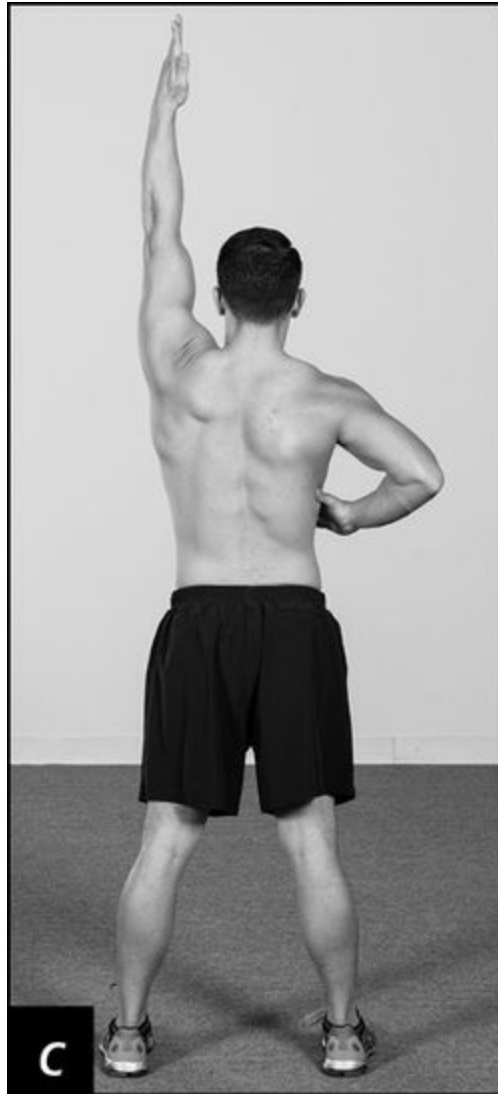
The scoliosis test reveals whether scoliosis (a) is correctable. The patient grabs an overhead bar. As the patient hangs supporting his full body weight with the arms, observe the spine curves. If the curves straighten under the traction load from the added body weight, the spine has the potential to respond to exercise therapy (b). Schroth therapeutic exercise is then administered (c). (Special condition exercises are beyond the scope of this

book, although one is shown as an example.)

True scoliosis is sometimes confused with temporary conditions. False signs suggesting a temporary scoliosis include unilateral psoas contraction, which gives an appearance of an aberrant curve. Some clinicians falsely interpret this as a short leg, perform an adjustment, and declare that they have corrected the short leg—in fact, they released the psoas. Obviously, a psoas contraction that disturbs normal hip and spine mechanics requires targeted treatment to the psoas, of which there are several appropriate soft tissue release techniques.







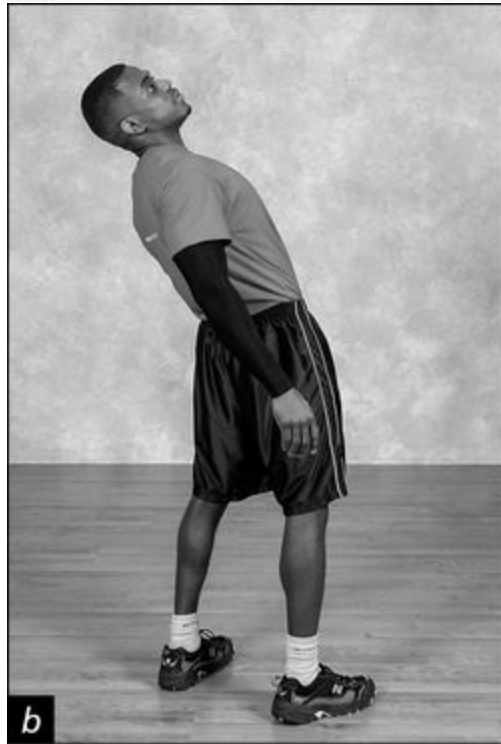
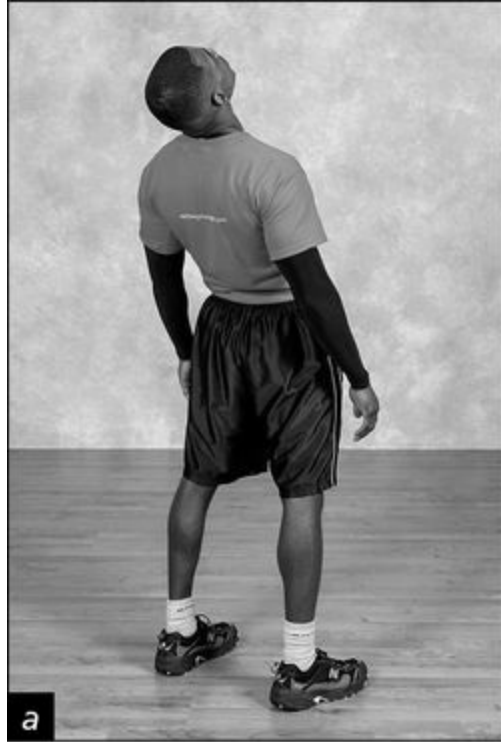
An example of a Schroth exercise for a contained lumbothoracic curve with a level pelvis and no substantially misshapen vertebrae is as follows. The hand contacts the rib cage on the convex side of the curve and applies pressure. The patient raises the arm on the concave side and pushes vertically. During the effort, the patient takes a deep breath, imagining the lungs filling with air on the concave side. The lung on the concave side is usually smaller.

Standing Extension Tests

Classically, extension postures have been thought to indicate posterior element or neural arch damage. However, as with disc pathology, this exacerbation provides only circumstantial evidence. The standing extension provocation tests include the extension quadrant and the one-legged stork

test. Both postures load the facet capsules, compress the facet articular interface, and bend the pars and lamina. They also close down the size of the lateral nerve root foramen and tension the anterior longitudinal ligament and anterior annulus. Understanding the interplay between these tissues helps in understanding the signs during the standing extension postures. For example, sagittal plane extension may cause central pain that is further exacerbated by lateral bending to the right, then twisting to drop the right shoulder back. Shown in the photos are (a) standing extension, (b) the extension quadrant test, (c) the one-legged stork test.







The main implication, as with many provocation tests, is that the reaction to extension postures provides circumstantial evidence. When combined with other evidence, the impression and working diagnostic hypothesis are strengthened or weakened. Relief with prone extension appears to be a strong indicator of discogenic pain from posterior annulus damage, but exacerbation of pain requires more follow-up testing of facets, the neural arch, and so on. If the extension pain is reduced or eliminated during the stork test, this adds more weight to a diagnosis of joint instability because the stork test adds stiffness and stability to the joints in the frontal plane. Nonetheless, in terms of designing therapy, the pain exacerbator (an extended spine in this case) has been identified and should probably be avoided.

Neural Tests

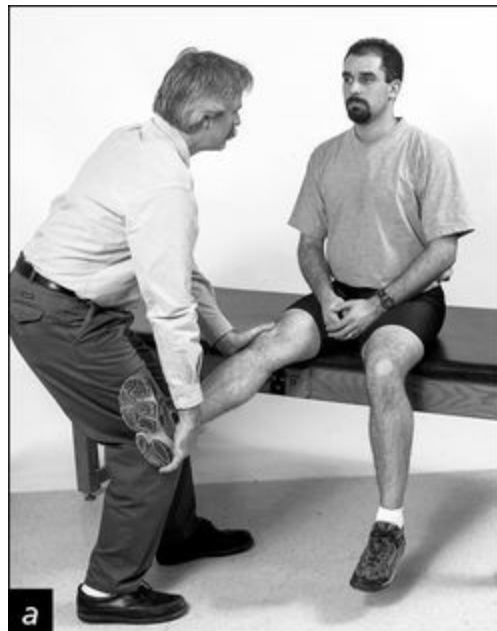
Pain is sometimes wrongly attributed to tight muscles when the genesis is actually an irritated nerve. Proper diagnosis is an important issue because the treatment of tight muscles usually makes an irritated nerve worse. The clinical techniques discussed here will help you enhance patient outcomes.

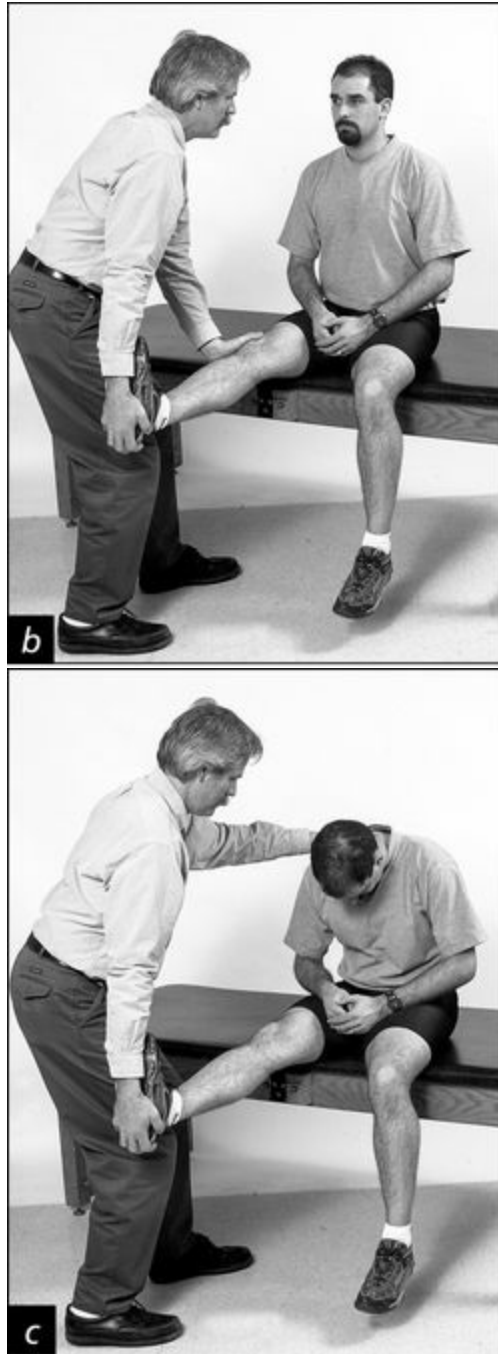
Sitting Slump Test

Several forms of the sitting slump test can be used to elicit radiating sciatic symptoms. These are designed to tense the sciatic nerve and irritate the lumbar nerve roots. Typically, the patient sits on the table or chair and then slumps or slouches. This test may be performed at the same time as the seated compression tests, because the load may work with the various postures to elicit pain. The intention is to progressively increase the nerve tension and define the trigger levels and conditions.

The test is as follows:

- (a) The leg is extended at the knee, tensing the nerve from below the lumbar spine.
- (b) Foot flexion can be added to further tense the nerve.
- (c) The nerve can be further tensed from above the lumbar region with the addition of cervical spine flexion.





Photos from Stuart McGill

If the tension at any of these stages causes symptoms, progressive testing should be stopped. The patient may be a candidate for nerve flossing (described in chapter 10). Certainly, forms of flexion are contraindicated for these patients during stretching and exercise.

If flexing the neck does not increase the pain, then the sciatic nerve is not the tissue of interest and you can begin assessing the sacroiliac (SI) joints, the

hamstrings, piriformis, and other muscle-based syndromes, ending with the hip. However, the slump test, in this form, is limited—a more insightful nerve tensioning test (the supine passive leg raise test) is described following the description of the Fajersztajn test and the neural underhook.

Fajersztajn Test

The Fajersztajn test should always be performed on both legs even when the sciatica is unilateral. Raising the well leg is needed to perform the well leg-raising test of Fajersztajn (DePalma and Rothman, 1970). Raising the well leg tensions the nerve root on the well side while causing tensions centrally along the midline of the cauda equina and to the nerve roots on the opposite side (a). Sometimes pain is provoked with simultaneous cervical flexion (b). Pain in the symptomatic side (side not raised) is an organic sign of disc lesion, usually a more central lesion. It is not a sign of malingering, as some have suggested.

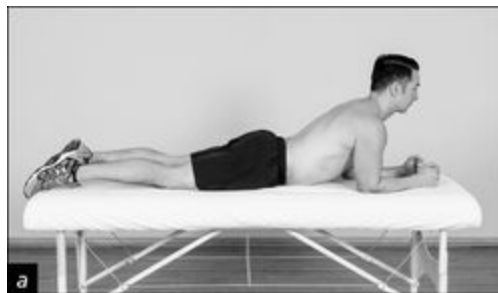




Photos from Stuart McGill

Neural Underhook

During neural tensioning tests, cervical flexion usually increases radiating lower limb symptoms—but not always. Sometimes cervical flexion relieves symptoms and extension increases symptoms. This sign usually indicates an underhook on the nerve root as the whole spinal cord slides caudally into the disc bulge. It also accounts for some patients who walk with their heads down in chronic cervical flexion, vexing clinicians who cannot understand why they do not stand upright. With this condition, the approach is usually to address the disc bulge with prone static McKenzie postures, and by avoiding lumbar flexion during daily activity. Over time, the ability to obtain an aligned, upright, pain-free posture should improve. The underhook also explains why some nerve flossing procedures require modified cervical postures and movements. Begin prone (a) with cervical extension, then only flex the lumbar spine (b).





Supine Passive Leg Raise Test (Sciatic Roots)

Technique while performing the leg raise test is very important. Both neural tension and muscular tension are changed with raising the leg, as in the sitting slump test. Thus, during the leg raise, some patients report pain that originates from muscle tension. To sort out the difference, always place one hand behind the knee so that the fingers can palpate the tension in the hamstring tendons. The patient's description of pain also assists in the impression of whether the pain is from neural or muscular tension—neural pain travels beyond the borders of the muscle (perhaps follow the sciatic tract in the buttock and below the knee). Internal rotation of the leg adds more tension to the sciatic tract.

If neural tension is present, follow up by determining whether the nerve root is free to floss. With the patient remaining in the supine posture, raise the pain-generating leg to the point of pain. The patient then raises the head with cervical flexion, and if that exacerbates the leg pain, this indicates that tension of the neural tract from above exacerbates the pain. Then lower the leg slightly. Alleviation of the pain with this maneuver establishes that releasing tension in the neural tract from below (while the tension from above remains) allows the nerve root to relax. Then returning the leg to the former position that generated pain and returning the head to the rested position on the table should reduce the pain. If this occurs, you will have established that the nerve root is flossing and not adhered. An adhered or pinched nerve root will not show different pain patterns from the coordinated cervical motion described here. Although we have been able to break nerve root adhesions (postsurgical) in the past with flossing regimens, the risk of creating exacerbated sciatica is large and requires strict guidance, which is discussed in chapter 10.





Shown in the photos is (a) the leg raised to the point of pain provocation and (b) cervical spine flexion added. Does this make the pain worse? Then (c) lower the leg. Does this reduce symptoms? If so, the nerve is flossing. Note the technique of cradling the leg in the crook of your arm so that your fingers can palpate the hamstring tendons.

While performing the supine passive leg raise, monitor muscle tension by using the fingers to palpate the hamstrings. (a) Raise the leg to the point of pain provocation. (b) Then have the patient flex the cervical spine to further tension the spinal cord from above the lumbar roots. If the pain is exacerbated either in the back or along the sciatic tract in the lower extremity, neural tension is considered the culprit. (c) Then lower the leg. If the pain is reduced, this is a clear indication that the nerve root is not adhered.

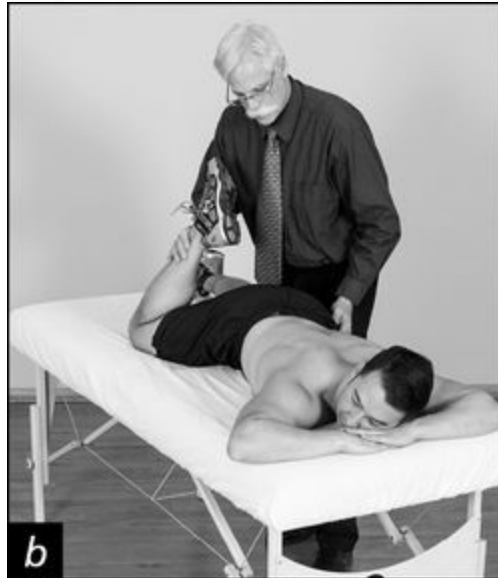
Tests for Femoral Nerve Roots

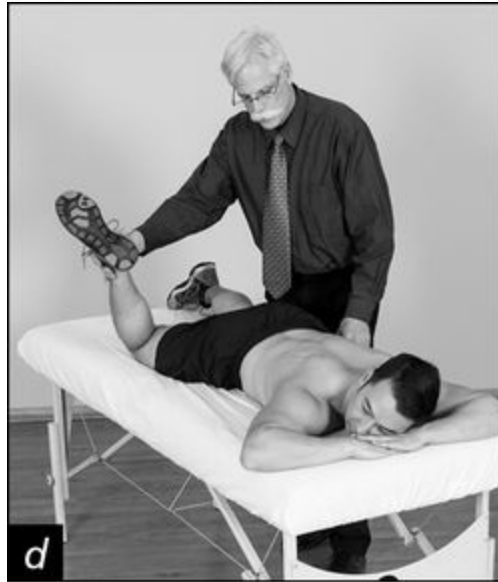
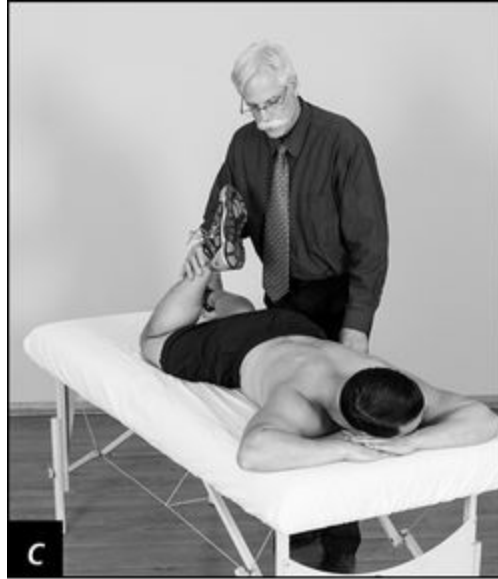
The upper lumbar spine (L1, L2, and L3) roots form the femoral nerve tract that is tensioned with hip extension and knee flexion. Pain-causing tension is tested in a prone posture. The clinician raises the lower leg into full knee flexion; then internally and externally rotate it. Femoral nerve roots are

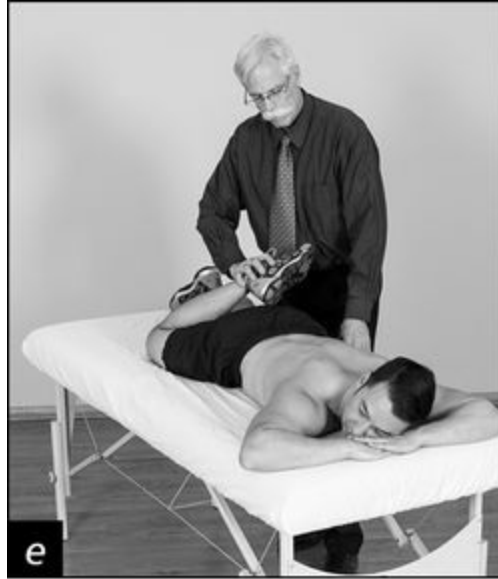
tensioned as the knee is flexed and the thigh is externally rotated. The pelvis rolling away from the knee being flexed is known as Ely's sign, which indicates either muscle or nerve tension. Follow up with a standing test to thrust the pelvis forward to tension the anterior hip capsule, while changing neural tract tensions to differentiate between muscle and nerve tension. Neural tensions are adjusted with cervical posture; frontal plane spine bending is probed with shoulder elevation.



Occasionally, this test may provoke back pain or radiating symptoms down the anterior thigh. If this occurs, manually stabilize the pelvis to see whether lumbar or SI joint instability is part of the pain trigger. Use cues for stiffening selected muscle groups to sort out pain-reducing potential.









The patient lies prone (a) as the clinician flexes the knee and asks the patient about pain or nerve crawling in the anterior thigh or back (b). The patient's head is rotated to the other side to see if perceptions change (c). The hip is internally (d) and externally rotated (e), then the clinician manually stabilizes the pelvis to probe whether this added control and stiffness changes the pain pattern (f, g).



Nerve tensions can be caused by pulls from either end. Here is an

example where cervical posture affects femoral nerve tension and corresponding symptoms. The femoral nerve is tensioned by the clinician (a) and as an assistant flexes and extends (b) the cervical spine to reveal the mechanism of posture-driven pain.

Differential Neural Tests and Implications

Cervical nerve roots are different from lumbar nerve roots in that they have ligamentous connections to vertebrae making them susceptible to whiplash, unlike lumbar roots. In addition, cervical roots have separate pain and sensory roots, whereas these are not separate in the lumbar roots (LaBan, 2008). This means that pain and weakness are more correlated with function associated with the lumbar roots.

It is possible for higher neural pathology to influence lumbar symptoms. We have seen posttrauma patients who experienced spinal cord shock and the subsequent death of cells within the cord. They were labeled as having bizarre behavior by the compensation board clinicians until we found the damage in the thoracic cord. This damage leaves deficits in motor engrams normally encoded in this region of the spine. For example, one person forgot how to cross his legs. Many weeks were required to recode this movement into the movement pattern repertoire. Another person forgot how to coordinate movement patterns to rise from a chair. This needed to be patiently relearned. Many neurologists seem unaware of this aspect of cord shock and movement behavior. Another thing to keep in mind is that EMG test results are not always truthful during the few weeks following trauma or the onset of acute back pain.

Foot drop is another condition suggesting pathology in the L4 nerve root. Jeon et al. (2013) introduced a test to differentiate a lumbar genesis from a peroneal nerve genesis. Weakness when testing hip abduction suggested a lumbar pathology. These researchers then followed up with Trendelenburg and gluteal muscle tests (shown as the exercises of the side-lying and clamshell tests) to determine whether an L5 root pathology existed.

The typical tests of myotomes are also helpful for determining pathological levels (e.g., great toe dorsiflexion, ankle plantar and dorsal function).

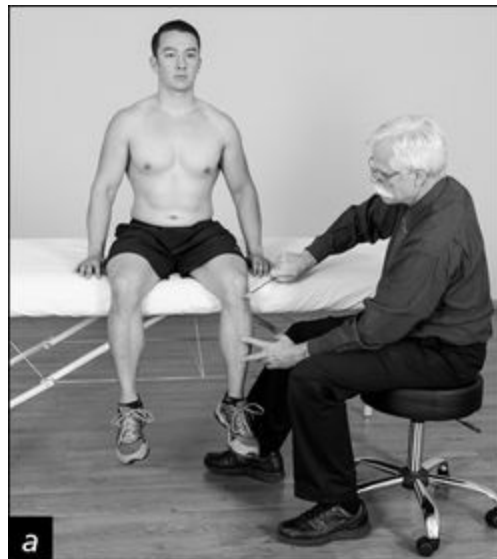
Reflex Testing to Find the Best Postures for Nerve

Mobilization

Peripheral reflexes such as the patellar reflex or the Achilles reflex are modulated by lumbar root tension or compromise. However, the actual nerve root tension that is modulated by posture is rarely considered. For example, patients with a diminished patellar reflex during the typical upright sitting posture are recorded as compromised. Yet in some of these patients, the reflex is restored when the posture is changed to reduce the neural tension. In this case, a logical posture is to have the patient lie back. A stronger reflex measured in this new posture establishes that the reflex is postural, and therefore biomechanically modulated.

Test for Posturally Modulated Reflex Diminution

Reflex intensity is evaluated as a function of neural tension. Adjust the seated posture of the patient to modulate the neural tension to determine whether the reflexes are modulated—for example, (a) with a neutral spine; (b) in a flexed posture, causing neural tension; or (c) in an extended posture, slackening the neural tract (or other postures between the two). If a stronger reflex response is generated in posture b, then this is adopted during flossing routines.





Testing for Lumbar Joint Shear Stability

Chapters 3 and 4 offer a well-developed discussion of the role of the lumbar extensors (*longissimus thoracis* and *iliocostalis lumborum*) in supporting anteroposterior shear of the vertebral motion segments. These muscles stiffen against unstable motion in the shear plane. In fact, good stabilization exercises must involve these muscles. A simple manual test for lumbar joint shear stability is presented here. If the result is positive (i.e., shear instability provokes pain), the patient could benefit from reducing the aberrant shear motion and will do well with stabilization exercises that

groove cocontracting motor patterns of the shear musculature (Hicks et al., 2005).

Manual Testing for Lumbar Joint Shear Stability

The patient lies prone on a table with the legs over the edge and the feet on the floor (a). The person must relax the torso musculature. Correct positioning is essential for true results. Too much spine flexion causes pain immediately in some patients (b). You then apply direct force downward onto each spinous process in turn (starting at the sacrum, then L5, L4, L3, etc.). The force applied should be no more than 1 kg (2 lb), and is applied in an oscillatory way to rattle the segment. Unstable segments are identified when either the person reports pain or you feel actual shear displacement. The patient's reporting of segment-specific pain, in this case, is given more consideration. Then ask the patient to slightly raise the legs off the floor to contract the back extensors. You once again apply force on each spinous process, using the same technique as before. By virtue of their lines of action, the lumbar extensors will reduce shearing instability, if present. If pain is present in the resting position but then disappears or subsides with the active cocontraction, the test is positive. Patients who indicate more pain during active contraction may be compression intolerant. Or they may be trying to extend their legs with lumbar extension rather than hip extension. Correction of this aberrant pattern may make the test more precise. More technique adjustments enhance insight into the pain exacerbator or reliever. Try more hip extension with a conscious gluteal squeeze and find the sweet spot (e).



This test proves conclusively that activating the extensors stabilizes the shear instability and eliminates the pain. Now the trick is to incorporate these extensor motor patterns into exercise prescriptions to carry over to daily activities.







Locating where to apply force: The thumbs palpate the posterior–superior iliac spines (PSISs) (f) on a skeleton model. As in real life (g), the thumbs slide to the midline. This will either be the posterior spine of L5 or the first sacral tubercle (h).



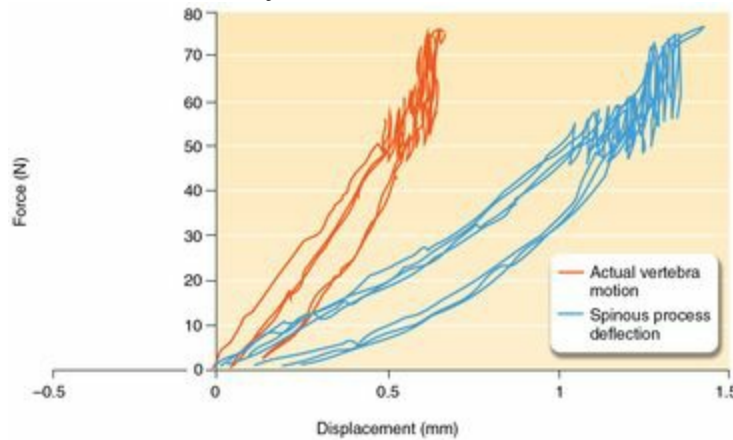


A Note on Motion Palpation

Some clinicians use their hands to feel specific vertebral motion during a whole-trunk motion such as flexion or twisting. When they detect a blocked or stiff-feeling segment, they often declare it pathological and make it the target of mobilizing treatment. Clinicians rarely, however, consider the flexibility in the spinous process and bony neural arch noted in chapter 3. Two of my PhD students questioned whether the asymmetrical feel to a specific joint was true pathology or simply asymmetrical anatomy—perhaps asymmetrical facet joints (Ross, Bereznick, and McGill, 1999). By carefully documenting the motion occurring at the disc and in the spinous process in cadaveric specimens, as well as the applied loads (similar to clinical loads), they were able to quantify the effect of anatomy. The results demonstrated that over half of the motion felt by the clinician was actually the spinous process flexing and bending (see [figure 9.1](#)) and not vertebral body motion. Those joints that were anatomically asymmetrical had asymmetrical motion under an applied load from the clinician. Treating such a joint with

mobilization therapy would be fruitless.

Figure 9.1 These results were measured as clinicians applied lateral forces to the tip of the spinous process of cadaveric motion segments to mimic motion-palpation testing. Most of the motion measured in the spinous process, which is what the clinician feels, was the result of spinous process bending rather than disc motion. Moreover, asymmetrical anatomy resulted in asymmetrical motion when loaded to mimic the clinician's hand forces. This suggests that this test should not be the only test to determine the site of treatment.



Distinguishing Between Lumbar and Hip Problems

Pain in the buttock or radiating down the leg (or both) can have more than one source. For effective treatment, you must discover whether that source is in the lumbar spine or the SI joints, in various muscles, in tissues around the hip, or even in the hip joint itself.

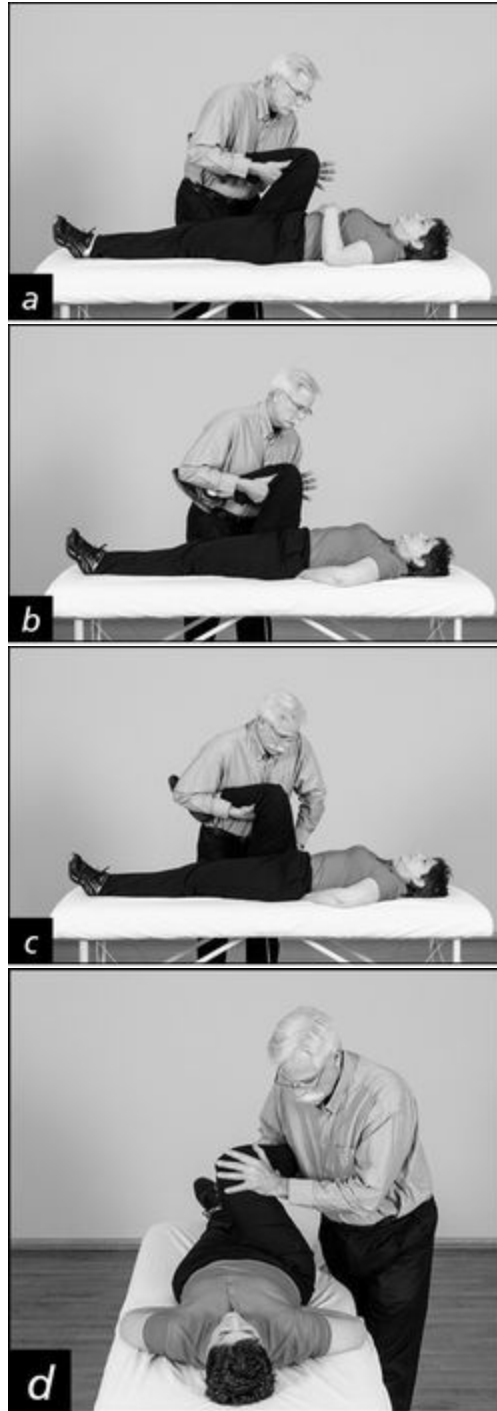
I am continually surprised at the number of people with back troubles who also have hip troubles (McGill and colleagues, 2003, noted a correlation between back troubles and hip pathology in factory workers). It is sometimes difficult to separate hip pain from lumbar trouble because deep buttock pain can be from a sciatic root (lumbar) or originate in the hip joint. Typically, medial and anterior thigh pain are indicators of hip pathology. If the nerve tension tests are negative and more hip investigation is indicated, then hip flexion and rotation tests should be performed.

Hip Flexion and Rotation Tests

Have the patient lie supine with the hip flexed. Rotate the patient's thigh segment into hip internal rotation (a). Stiffness or pain is a positive indicator of hip concerns. Hip joint pain is somewhat distinctive with radiation into the groin, or along the inguinal crease or down the anterior–medial thigh (or some combination of these). When more pressure is applied to the thigh challenging the hip joint in full flexion, the spine will also flex. When back pain occurs during hip flexion, stabilize the lumbar spine with the hand (c). Also have the patient place his hands under the lumbar spine to prevent flexion. If this causes back pain, stabilize the pelvis with your other hand so that no spine motion occurs and rotation is confined to the hip. If the back pain is gone, the spine was the source.

Then flex the patient's hip (d) while moving the knee in circular motions, scouring the acetabulum, but with the knee bent so as not to confuse any pain with nerve tension related to the back. Now externally rotate the thigh, testing the joint capsule (e) and stabilizing the pelvis to isolate the hip (f-h). Finally, the hip distracted (i) and compressed (j).





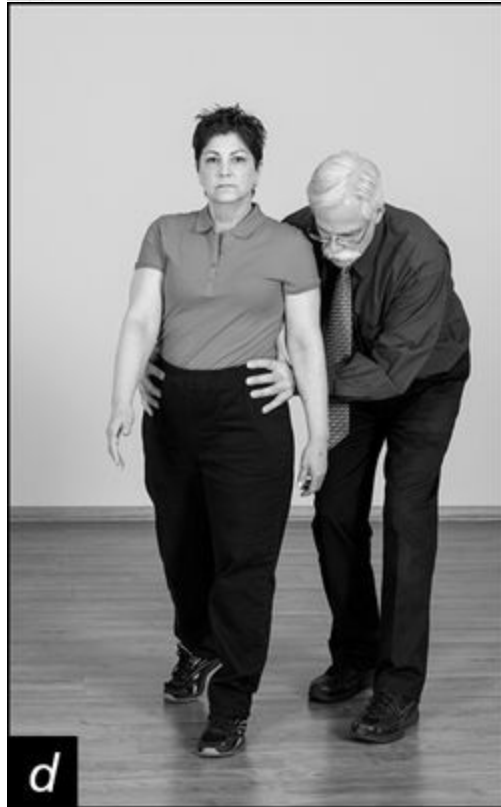




Tests for Sacroiliac Joint Pain

Young, Aprill, and Laslett (2003) noted that SI joint pain is rarely experienced along the midline, above the L5 level, and bilaterally. Rising from a chair tends to exacerbate the pain. Our clinical observations suggest that pain is provoked with lunges. Several tests challenge the SI joints and pelvic ring. The series begins with force applied down into the hip joint, followed by an uplifting force similar to that shown in photos i and j of the hip flexion and rotation test just described. Then the supine patient bridges the pelvis up and lowers the sacrum down onto your fist (see a and b). This may cause pain. Or you can perform rapid fist clenches to transfer sharp pulses into the pelvis via the SI joints. The clinician's other hand supports the lumbar spine, changing the amount of lordosis to probe whether this influences pain (c). This loads the SI joints in pulsating shear loads. Then the patient walks while you manually apply pressure to the iliac crests cracking the low half of the SI open (d, e). The patient then repeats the walk with manual pressure applied to the greater trochanters, cracking open the lower half of the SI joints (f, g). These tests help define the nature of the SI pain if it present. If no pain is provoked, the SI joints can be ruled out.







Manual Iliac Crest Compression Test

Continue probing the pelvic ring to learn more about the cause of the pain and what may help. If manual compression relieves the pain, try cuing

abdominal wall stiffness while the patient walks to see whether this reduces pain. If any of these strategies relieves pain, it is a strong indicator of pelvic ring instability. The exercises prescribed for initial lumbar stabilization (chapter 10) also stiffen and stabilize the pelvis. Also, probe the patient to find the cause. Many report recent changes in their training such as adding many more loaded split squats. Too many of these exercises can loosen the SI joints. Postpartum women, particularly those who have returned to vigorous activity too soon, or those who have sustained a high-energy trauma such as a fall or motor vehicle accident, may present with unstable pelvic rings and associated back or pelvic pain.

